

A quick review of Bayesian methodology

Statistical analyses in the cognitive sciences often pursue two goals: to estimate parameters and their uncertainties, and to test hypotheses. Both of these goals can be achieved using Bayesian data analysis. Bayesian analyses focus on an “observational” model $\mathcal{M}$, which specifies the probability density of the data y given the vector of model parameters Θ and the model $\mathcal{M}$, i.e., $p(y \mid \Theta, \mathcal{M})$, or by dropping the model, $p(y \mid \Theta)$. It is possible to use the observational model to simulate data, by selecting some model parameters Θ and drawing random samples for the data $\tilde{y}$. When the data is given (fixed, e.g., observed or simulated), then the observational model turns into a likelihood function: $p(y \mid \Theta) = L_y(\Theta)$; this can be used to estimate model parameters or to compute the evidence for the model relative to other models. To estimate parameters, in Bayesian data analyses the likelihood is complemented by the prior, written $p(\Theta)$, that defines the joint probability distribution of all the parameters of the model, which can encode domain expertise. Bayes’ rule specifies how to combine the likelihood and the prior to compute the posterior distribution of the model parameters $p(\Theta \mid y, \mathcal{M}_1)$:

$$p(\Theta \mid y, \mathcal{M}_1) = \frac{p(y \mid \Theta, \mathcal{M}_1)p(\Theta \mid \mathcal{M}_1)}{p(y \mid \mathcal{M}_1)} \quad (1)$$

Here, $p(y \mid \mathcal{M}_1)$ is a normalizing constant termed the “evidence” or “marginal likelihood”; it is the likelihood of the data y based on the model $\mathcal{M}_1$ averaging over the parameters Θ . The marginal likelihood can only be interpreted relative to other models. It is derived as $p(y \mid \mathcal{M}_1) = \int p(y \mid \Theta, \mathcal{M}_1)p(\Theta \mid \mathcal{M}_1) d\Theta$. There is a key role for priors in this computation. Priors play an important role in Bayesian inference since expressing the prior is a means of quantifying the researcher’s uncertainty about the model parameters. From a practical perspective, in Bayesian inference, priors can regularize inferences when the data do not strongly inform the likelihood functions. However, they will influence marginal likelihoods even when the data are strongly informative, and are thus even more crucial for Bayes

factors than for posterior distributions (Aitkin, 1991; Gelman et al., 2013; Grünwald, 2000; Kass & Raftery, 1995; Liu & Aitkin, 2008; Myung & Pitt, 1997; Ravenzwaaij & Wagenmakers, 2022; Tendeiro & Kiers, 2019; Vanpaemel, 2010).

For very simple models, posterior density functions can be computed analytically. However, for most interesting models this is not possible and we have to rely on approximation methods. One such approach is sampling methods such as Markov Chain Monte Carlo sampling (Gamerman & Lopes, 2006), which is an important method behind popular software implementing Bayesian analysis.

Inference over hypotheses. Bayes factors provide a way to compare any two model hypotheses (i.e., arbitrary hypotheses) against each other by comparing their marginal likelihoods (Betancourt, 2018; Kass & Raftery, 1995; Ly, Verhagen, & Wagenmakers, 2016; Schad et al., 2023). The Bayes factor tells us, given the data and the model priors, how much we need to update our relative belief between the two models.

To derive Bayes factors, we first compute the model posterior, i.e., the posterior probability for a model $\mathcal{M}_i$ given the data:

$$p(\mathcal{M}_i | y) = \frac{p(y | \mathcal{M}_i) \times p(\mathcal{M}_i)}{p(y | \mathcal{M}_1) \times p(\mathcal{M}_1) + p(y | \mathcal{M}_2) \times p(\mathcal{M}_2)} = \frac{p(y | \mathcal{M}_i) \times p(\mathcal{M}_i)}{p(y)}. \quad (2)$$

Here, $p(\mathcal{M}_i)$ is the prior probability for each model i . Based on the posterior model probability $p(\mathcal{M}_i | y)$, we can compute the model odds for one model over another as:

$$\frac{p(\mathcal{M}_1 | y)}{p(\mathcal{M}_2 | y)} = \frac{[p(y | \mathcal{M}_1) \times p(\mathcal{M}_1)]/p(y)}{[p(y | \mathcal{M}_2) \times p(\mathcal{M}_2)]/p(y)} = \frac{p(y | \mathcal{M}_1)}{p(y | \mathcal{M}_2)} \times \frac{p(\mathcal{M}_1)}{p(\mathcal{M}_2)}. \quad (3)$$

In words:

$$\textit{Posterior ratio} = \textit{Bayes factor} \times \textit{prior odds} . \quad (4)$$

The Bayes factor is thus a measure of relative evidence, the comparison of the predictive performance of one model ($\mathcal{M}_1$) against another one ($\mathcal{M}_2$). This comparison (BF_{12}) is a ratio of marginal likelihoods:

$$BF_{12} = \frac{p(y \mid \mathcal{M}_1)}{p(y \mid \mathcal{M}_2)} . \quad (5)$$

BF_{12} indicates the evidence that the data provide for $\mathcal{M}_1$ over $\mathcal{M}_2$; in other words, which of the two models is more likely to have generated the data, or the relative evidence that we have for $\mathcal{M}_1$ over $\mathcal{M}_2$. Bayes factor values larger than one indicate that $\mathcal{M}_1$ is more compatible with the data, values smaller than one indicate $\mathcal{M}_2$ is more compatible with the data, and values close to one indicate that both models are equally compatible with the data.

In the present work, we will consider the case of nested model comparison. Here, a null model hypothesizes that a model parameter is zero or absent (a point hypothesis). By contrast, an alternative model hypothesizes that the model parameter is present with some prior distribution and has some value different from exactly zero (a “general” hypothesis).

As mentioned above, for most interesting problems and models in cognitive science, Bayes factors cannot be computed analytically; approximations are needed. One major approach is to estimate Bayes factors based on posterior MCMC draws via bridge sampling (Bennett, 1976; Gronau et al., 2017; Meng & Wong, 1996), implemented in the R package `bridgesampling` (Gronau et al., 2020), which we use in the present work. We chose this package for the present study because it is widely used in cognitive science, and because it allows for straightforward application of marginal SBC. We did not use the Bayes factor package for analysis. Simulation-based calibration requires simulating from the parameter